

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Muthu Krishnan K/192511039 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Muthu Krishnan K

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.
6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.
7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.
8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.
9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation ; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis

Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
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		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File Size = 150 MB = $150 \times 10^6 = 150,000,000$ bytes
- Segment Size = 1500 bytes
- Data Link Overhead = 40 bytes/frame
- Bandwidth = 50 Mbps

Solution:

Number of Segments

= $150,000,000 / 1500$

= **100,000 segments**

Number of Frames

= **100,000 frames**

Total Overhead

= $100,000 \times 40$

= **4,000,000 bytes = 4 MB**

Total Data Transmitted

= $150,000,000 + 4,000,000$

= **154,000,000 bytes**

Convert to bits

= $154,000,000 \times 8$

= **1,232,000,000 bits**

Transmission Time

= $1,232,000,000 / 50,000,000$

= **24.64 seconds**

Answer:

- Total Segments = **100,000**
- Total Frames = **100,000**
- Data Link Overhead = **4 MB**

- Transmission Time = **24.64 seconds**

Explanation:

The Transport Layer performs segmentation, sequencing, flow control, and error recovery. The Data Link Layer performs framing, error detection, and reliable node-to-node communication.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Window Size = 6 frames
- Total Frames = 48

Solution:

Number of Windows

= $48 / 6$

= **8 windows**

One frame lost per window

Retransmissions

= 8×1

= **8 retransmissions**

Answer:

- Number of Windows = **8**
- Total Retransmissions = **8**

Explanation:

Flow control prevents the sender from overwhelming the receiver. Sliding window protocols improve network utilization and reliability through acknowledgments and retransmissions.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- Packet Size = 12,000 bytes
- MTU = 1500 bytes
- IP Header = 20 bytes

Solution:

Maximum Data per Fragment

$$= 1500 - 20$$

= 1480 bytes

Number of Fragments

$$= 12,000 / 1480$$

$$= 8.11$$

= **9 fragments**

Header Overhead

$$= 9 \times 20$$

= **180 bytes**

Answer:

- Number of Fragments = 9
- Header Overhead = **180 bytes**

Explanation:

The Network Layer performs addressing, routing, fragmentation, and reassembly to ensure successful packet delivery.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Solution:

Number of Windows

$$= 48 / 6$$

= **8 windows**

Retransmissions

$$= 8 \times 1$$

= **8 retransmissions**

Answer:

- Number of Windows = 8
- Retransmissions = 8

Explanation:

Flow control avoids congestion and improves communication efficiency by regulating transmission speed.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the

importance of synchronization in the Session Layer.

Given:

- File Size = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Solution:

Checkpoints Before Failure

$$= 390 / 60$$

$$= 6.5$$

= **6 checkpoints**

Last checkpoint

$$= 6 \times 60$$

$$= 360 \text{ MB}$$

Retransmission Needed

$$= 390 - 360$$

$$= \mathbf{30 \text{ MB}}$$

Answer:

- Checkpoints Created = **6**
- Data Retransmitted = **30 MB**

Explanation:

Synchronization checkpoints enable communication recovery from the last saved state rather than retransmitting the entire file.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original Size = 900 MB
- Compression = 40%
- Encryption Overhead = 5%

Solution:

Compressed Size

$$= 900 \times 0.60$$

$$= \mathbf{540 \text{ MB}}$$

Encrypted Size

$$= 540 \times 1.05$$

$$= \mathbf{567 \text{ MB}}$$

Reduction

$$= ((900 - 567) / 900) \times 100$$

$$= \mathbf{37\%}$$

Answer:

- Compressed Size = **540 MB**
- Encrypted Size = **567 MB**
- Percentage Reduction = **37%**

Explanation:

The Presentation Layer handles compression, encryption, and data format conversion to improve efficiency and security.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Data = 3 GB = 3000 MB
- Request Size = 3 MB
- Throughput = 120 Mbps

Solution:

Number of Requests

$$= 3000 / 3$$

$$= \mathbf{1000 \text{ requests}}$$

Total Data in Bits

$$= 3000 \times 8$$

$$= 24,000 \text{ Mb}$$

Transmission Time

$$= 24,000 / 120$$

$$= \mathbf{200 \text{ seconds}}$$

Answer:

- Requests Generated = **1000**
- Transmission Time = **200 seconds**

Explanation:

The Application Layer provides services such as FTP, HTTP, email, authentication, and user interaction.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the

transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Solution:

Processing Delay
 $= 4 + 3 + 5 + 2 + 1$
 $= 15 \text{ ms}$
Total Delay
 $= 15 + 18 + 12$
 $= \mathbf{45 \text{ ms}}$

Answer:

- Total End-to-End Delay = **45 ms**

Explanation:

Each layer performs specific functions such as application services, segmentation, routing, framing, and signal transmission.

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful Data = 80 MB = 80,000,000 bytes
- Frame Size = 1600 bytes
- Overhead = 80 bytes

Solution:

Useful Data per Frame
 $= 1600 - 80$
 $= 1520 \text{ bytes}$
Frames Required
 $= 80,000,000 / 1520$
 $\approx \mathbf{52,632 \text{ frames}}$
Total Overhead
 $= 52,632 \times 80$
 $= \mathbf{4,210,560 \text{ bytes}}$
Efficiency
 $= (1520 / 1600) \times 100$
 $= \mathbf{95\%}$

Answer:

- Frames Required = **52,632**
- Total Overhead = **4,210,560 bytes**

- Efficiency = **95%**

Explanation:

Protocol overhead reduces usable bandwidth and increases transmission cost.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original Data = 240 MB
- Compression = 30%
- Segment Size = 1400 bytes
- Overhead = 60 bytes/frame

Solution:

Compressed Size

$$= 240 \times 0.70$$

= **168 MB**

Compressed Data

$$= 168,000,000 \text{ bytes}$$

Number of Segments

$$= 168,000,000 / 1400$$

= **120,000 segments**

Protocol Overhead

$$= 120,000 \times 60$$

= **7,200,000 bytes**

Final Data Transmitted

$$= 168,000,000 + 7,200,000$$

= **175,200,000 bytes**

= **175.2 MB**

Answer:

- Compressed File Size = **168 MB**
- Number of Segments = **120,000**
- Protocol Overhead = **7.2 MB**
- Final Data Transmitted = **175.2 MB**

Explanation:

The Presentation Layer compresses and encrypts data, the Transport Layer performs segmentation and reliability functions, the Data Link Layer provides framing and error detection, and the Physical Layer transmits bits across the network.